



**DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING LAB MANUAL**

CS23432 – Software Construction

(REGULATION 2023)

**RAJALAKSHMI ENGINEERING COLLEGE
Thandalam, Chennai-602015**

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Year / Branch / Section: II - IT Semester: IV

Academic Year: 2025-2026

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10.	22/04/2026	GitHub: Project Structure & Naming Conventions.

EXPNO:1	AZURE DEVOPS ENVIRONMENT SEUP
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Aim:

To set up and access the Azure DevOps environment by creating an organization through the Azure portal for developing the **Online Quiz System** project.

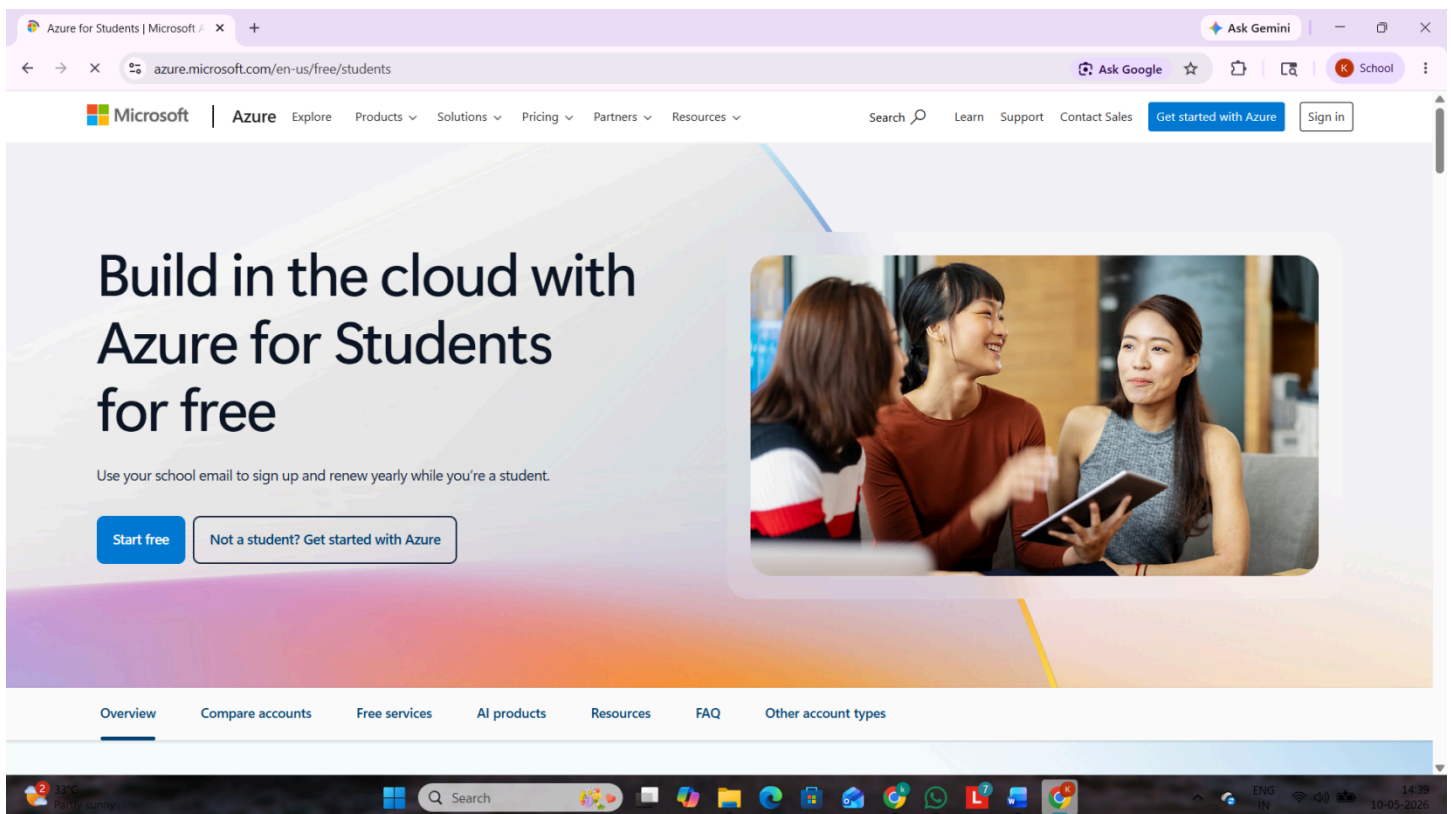
INSTALLATION

1. Open your web browser and go to the Azure website:

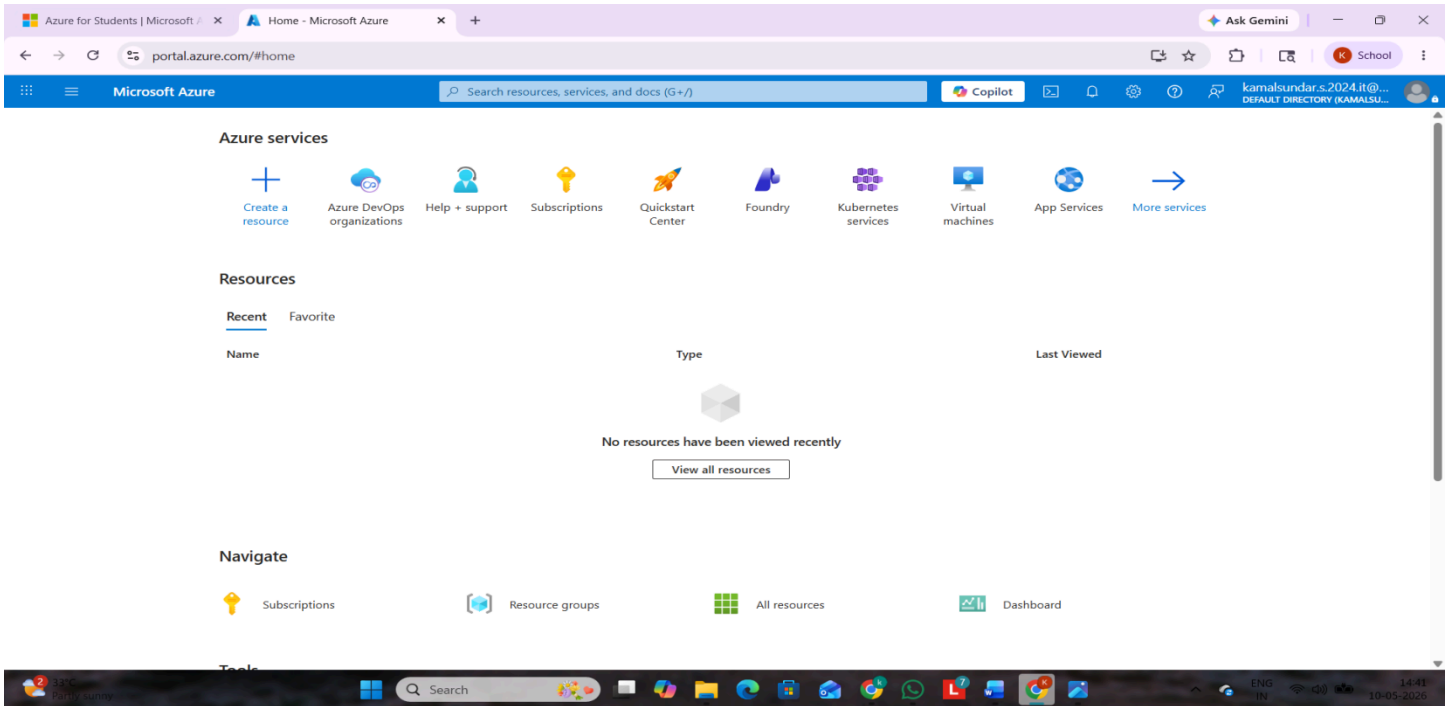
<https://azure.microsoft.com/en-us/get-started/azure-portal>.

Sign in using your Microsoft account credentials.

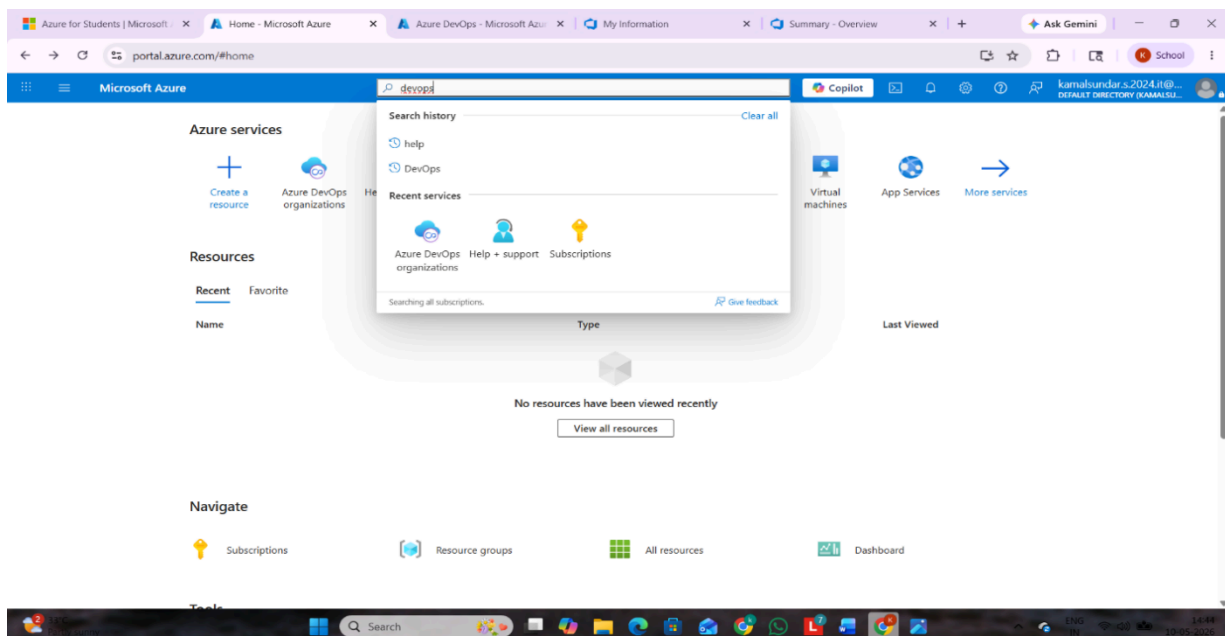
If you don't have a Microsoft account, you can create one [here: https://signup.live.com/?lic=1](https://signup.live.com/?lic=1)



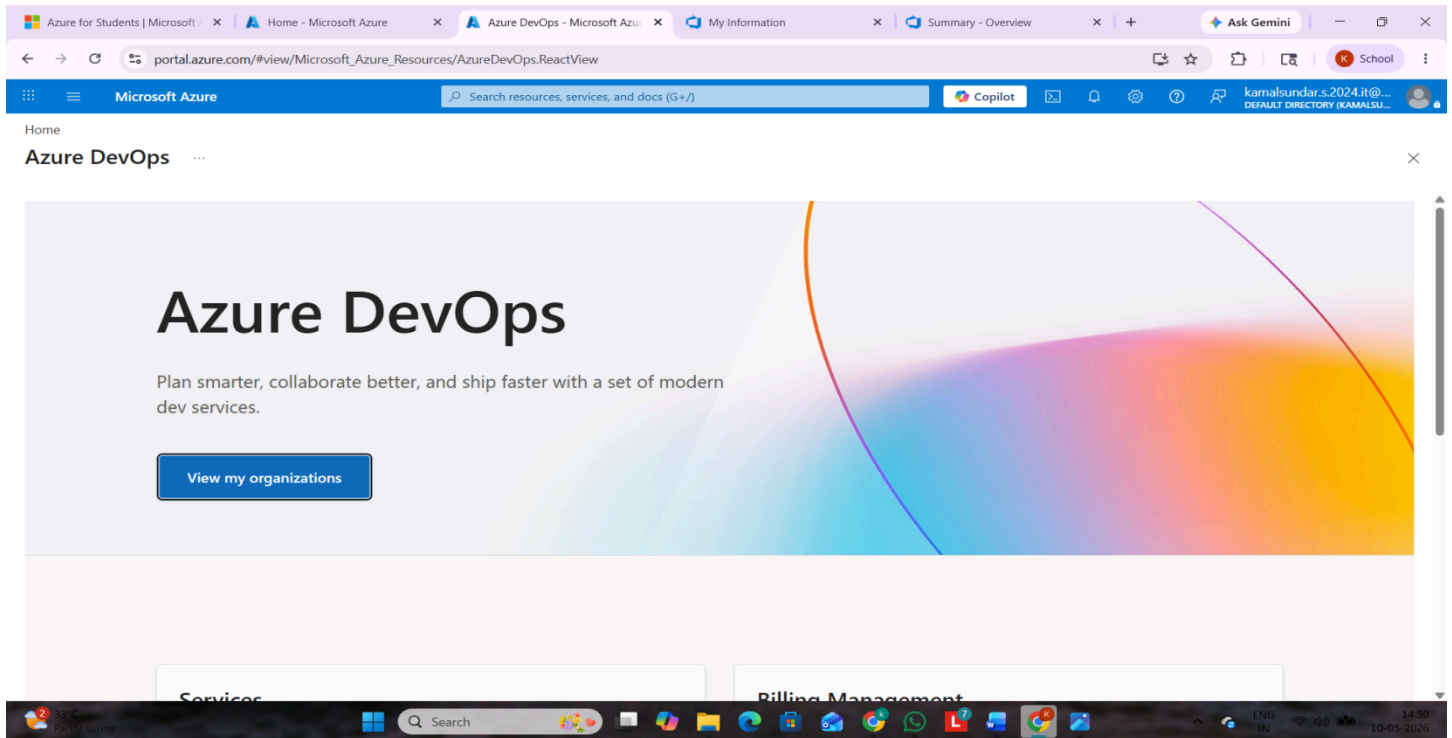
2. Azure home page



3. Open DevOps environment in the Azure platform by typing *Azure DevOps Organizations* in the search bar.



- Click on the *My Azure DevOps Organization* link and create an organization and you should be taken to the Azure DevOps Organization Home page.



Result:

Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal for the Online Quiz System project.

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EXPNO:2 C r e a	AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT
-----------------------------	---

Aim:

To set up an Azure DevOps project for efficient collaboration and agile work management for the Online Quiz System.

1. Create An Azure Account

aez.dev.azure.com/signup?acquisitionId=294d444a-d1b9-4123-89e1-7dc1cad70db9&campaign=o-msft-profile-service_attach&mkt=en-US

Azure DevOps
geevan.jv.2024.it@rajalakshmi.edu.in [Switch directory](#)

Almost done...

Name your Azure DevOps organization *

dev.azure.com/ geevanjv2024it

We'll host your projects in

India

Select an Azure subscription for billing [🔗](#)

Azure for Students (ab52830a-9a0f-42e5-b442-...

You can also [create a new Azure subscription](#)

Continue

2. Create the First Project in Your Organization

a. After the organization is set up, you'll need to create your first **project**. This is where you'll begin to manage code, pipelines, work items, and more.

b. On the organization's **Home page**, click on the **New Project** button.

c. Enter the project name, description, and visibility options:

Name: Choose a name for the project (e.g., **LMS**).

Description: Optionally, add a description to provide more context about the project.

Visibility: Choose whether you want the project to be **Private** (accessible only to those invited) or **Public** (accessible to anyone).

d. Once you've filled out the details, click **Create** to set up your first project.

Create new project

×

Project name *

Weather Apps

Description

Visibility

Private

Only people you give access to will be able to view this. Want to create a public project? [Try GitHub](#)

^ Advanced

Version control

Git

▼

Work item process

Agile

▼

Cancel

Create

3.Once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.

This screenshot shows the 'Azure DevOps Organizations' page for a user named Kamal Sundar. The page is divided into two main sections. On the left, the user's profile is displayed, including their name, email address (kamalsundar.s.2024.it@rajalakshmi.edu.in), and a Microsoft account dropdown menu. Below the profile, there is a section for 'Visual Studio Dev Essentials'. On the right, the 'Azure DevOps Organizations' section shows the organization 'dev.azure.com/harshithkumars2024it' with a 'Create new organization' button. Under the 'Projects' tab, the 'Online Quizz' project is listed. To the right of the project, there are 'Actions' such as 'Open in Visual Studio', 'Manage security', 'Browse extensions', and 'Leave'.

This screenshot shows the 'Online Quizz' project dashboard in Azure DevOps. The dashboard is divided into several sections. On the left, there is a sidebar with navigation links for 'Online Quizz', 'Overview', 'Summary', 'Dashboards', 'Wiki', 'Boards', 'Repos', 'Pipelines', 'Test Plans', and 'Artifacts'. The main content area is titled 'Online Quizz' and contains an 'About this project' section. This section describes the project as a scalable and user-friendly online quiz platform designed to facilitate interactive learning and assessment. It mentions that the development and deployment process is managed using Microsoft Azure DevOps, enabling efficient collaboration, version control, continuous integration, and continuous deployment (CI/CD). It also states that the platform is designed with a responsive user interface and a secure backend to handle user data and quiz content. On the right side of the dashboard, there are 'Project stats' for the last 7 days, showing 0 work items created, 0 work items completed, 0 pull requests opened, and 4 commits by 2 authors. Below the stats, there is a 'Members' section showing 4 members: GV, HS, HA, and KS.

4. Project dashboard

5. To manage user stories:

a. From the **left-hand navigation menu**, click on **Boards**. This will take you to the main **Boards** page, where you can manage work items, backlogs, and sprints.

b. On the **work items** page, you'll see the option to **Add a work item** at the top. Alternatively, you can find a **+** button or **Add New Work Item** depending on the view you're in. From the **Add a work item** dropdown, select **User Story**. This will open a form to enter details for the new User Story.

Azure DevOps harshithkumars2024it / Online Quizz / Boards / Backlogs

Search

Online Quizz

Overview

Boards

Work items

Boards

Backlogs

Sprints

Queries

Delivery Plans

Analytics views

Repos

Pipelines

Test Plans

Artifacts

Project settings

Online Quizz Team

New Work Item

View as Board

Column Options

Backlog

Analytics

	Order	Work Item Type	Title	State	Story...	Value Area
1		User Story	User Registration	Resolved	3	Business
2		User Story	Join Quiz	New	2	Business
3		User Story	Attempt Quiz	New	5	Business
4		User Story	View Results	New	3	Business

Stories

Planning

Drag and drop work items to include them in a sprint.

Online Quizz Team backlog

Iteration 1
Planned Effort: 13
4

Iteration 2
No work scheduled yet

Iteration 3
No work scheduled yet

New Sprint

Azure DevOps harshithkumars2024it / Online Quizz / Boards / Backlogs

Search

Online Quizz

Overview

Boards

Work items

Boards

Backlogs

Sprints

Queries

Delivery Plans

Analytics views

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Pipelines

Test Plans

Artifacts

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New Work Item

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4		User Story	View Results	New	3	Business

Stories

Planning

Drag and drop work items to include them in a sprint.

Online Quizz Team backlog

Iteration 1
Planned Effort: 13
4

Iteration 2
No work scheduled yet

Iteration 3
No work scheduled yet

New Sprint

Microsoft Sign out

Harshith Kumar Su...
harshithkumar.s.2024.it@rajal...
[My Microsoft account](#)
[Switch directory](#)

Sign in with a different account

Result:

Successfully created an Azure DevOps project with user story management and setup for the Online Quiz System.

EXP NO: 3

SETTING UP EPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING

Aim:

To create epics, features, user stories, and tasks for the Online Quiz System project planning.

Create Epic, Features, User Stories, Task

The screenshot shows the Azure DevOps interface for the 'Online Quiz Team'. The 'Backlog' view is active, displaying a list of work items. The table below represents the data shown in the backlog:

Order	Work Item Type	Title	State	Story...	Value Area
1	User Story	User Registration	Resolved	3	Business
2	User Story	Join Quiz	New	2	Business
3	User Story	Attempt Quiz	New	5	Business
4	User Story	View Results	New	3	Business

On the right, the 'Planning' sidebar shows the team backlog and three iterations (Iteration 1, 2, 3) with their planned effort and current work count.

1.Fill in Epics

The screenshot shows the 'Work items' view for an Epic titled 'Quiz Gamification System'. The work item is created by Harshith Kumar Suresh. The details are as follows:

- Description:** A system that enhances user engagement by adding game-like elements to quizzes. Users can earn points, badges, and rewards based on their performance, encouraging continuous participation and improvement.
- Discussion:** A comment box is present for adding comments.
- Planning:**
 - Priority: 2
 - Risk: 2 - Medium
 - Effort: 3
 - Business Value: 80
 - Time Criticality: 60
 - Start Date: 3/1/2026 12:00 AM
 - Target Date: 6/10/2026 12:00 AM
- Deployment:** A section for tracking releases, with a link to 'Releases'.
- Development:** A section for linking to Azure Repos, with a link to 'commit, pull request or branch'.

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1.Fill in Features

The screenshot shows the Azure DevOps interface for a project named 'Online Quiz'. The left sidebar contains navigation options: Overview, Boards, Work items, Backlogs, Sprints, Queries, Delivery Plans, Analytics views, Repos, Pipelines, Test Plans, and Artifacts. The main area displays a 'Feature' item titled '12 Timer-Based Quiz' created by 'harish raghava A'. The item is in the 'Resolved' state. The description states: 'This feature enforces a time limit for each quiz or question. A countdown timer is displayed to users, ensuring the quiz runs within a fixed duration. When time expires, the quiz is automatically submitted. The timing logic and triggers can be handled using Azure Functions for a timely and scalable execution.' The right sidebar shows 'Planning' details (Priority 1, Risk 2 - Medium, Effort 2, Business Value 3, Time Criticality 3, Start Date 3/1/2026 12:00 AM, Target Date 6/10/2026 12:00 AM) and 'Deployment' instructions. A 'Discussion' section at the bottom allows for comments.

2.Fill in User Story Details

The screenshot shows the Azure DevOps interface for the same 'Online Quiz' project. The main area displays a 'User Story' item titled '10 View Results' created by 'Geevan J V'. The item is in the 'New' state. The description states: 'As a user, I want to see my quiz results so that I can know my performance.' The 'Acceptance Criteria' section has a placeholder 'Click to add Acceptance Criteria.' The right sidebar shows 'Planning' details (Story Points 3, Priority 2, Risk 2 - Medium) and 'Classification' (Value area Business). A 'Discussion' section at the bottom allows for comments.

Result:

Thus, the creation of epics, features, user story and task has been created successfully.

EXPNO: 4	SPRINT PLANNING
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EXPNO: 4	SPRINT PLANNING
----------	-----------------

Aim:

To assign user stories to specific sprints for the Online Quiz System.

Sprint Planning

Sprint1

Azure DevOps

harshithkumars2024it

Online Quiz

Boards

Sprints

Queries

Delivery Plans

Analytics views

Repos

Pipelines

Test Plans

Artifacts

Project settings

Online Quiz

Overview

Boards

Work items

Boards

Backlogs

Sprints

Queries

Delivery Plans

Analytics views

Repos

Pipelines

Test Plans

Artifacts

Project settings

Online Quiz Team

Taskboard

Backlog

Capacity

Analytics

+ New Work Item

Column Options

Iteration 1

Person: All

7 User Registration

Resolved

Harshith Kumar Suresh

8 Join Quiz

New

Kamal Sundar

9 Attempt Quiz

New

harish raghava A

10 View Results

New

Geevan J V

No iteration dates

Set dates

Sprint 2

Azure DevOps harshithkumars2024it / Online Quiz / Boards / Sprints

Search

Online Quizz Team

+ New Work Item Column Options Create Query

Taskboard Backlog Capacity Analytics

Iteration 2

No iteration dates
Set dates

	Order	Title	State	Assigned To	Rema...
1		Join Quiz	New	Kamal Sundar	
2		Attempt Quiz	New	harish raghava A	
3		View Results	New	Geevan J V	

Sprint 3

Azure DevOps harshithkumars2024it / Online Quiz / Boards / Sprints

Search

Online Quizz Team

+ New Work Item Column Options Create Query

Taskboard Backlog Capacity Analytics

Iteration 3

No iteration dates
Set dates

	Order	Title	State	Assigned To	Rema...
1		View Results	New	Geevan J V	

Result:

The sprints were created successfully for the Online Quiz System.

EXP NO: 5

POKER ESTIMATION

Aim:

To create Poker Estimation for the user stories in the Online Quiz System project.

PokerEstimation

Azure DevOps harshithkumars2024it / Online Quizz / Boards / Work items

Search

Online Quizz

Recently updated | Back to Work Items

4 of 12

USER STORY 7

7 User Registration

Harshith Kumar Suresh | 2 Comments | Add Tag

Save | Follow | Settings | Refresh | Share | More

Updated by Harshith Kumar Suresh: 11m ago

Details | Clock | Comment | Like | Share

Description

As a user, I want to create an account so that I can access and participate in quizzes

Acceptance Criteria

- User should be able to register using college email ID
- Email must follow rajalakshmi.edu.in format
- Email should contain valid joining year (2023 to present)
- Password field should not be empty
- System should show error for invalid email
- System should prevent duplicate registrations
- Successful registration should display confirmation message

Discussion

Planning

Story Points: 3

Priority: 3

Risk: 2 - Medium

Classification

Value area: Business

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Collapse Classification section

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Result:

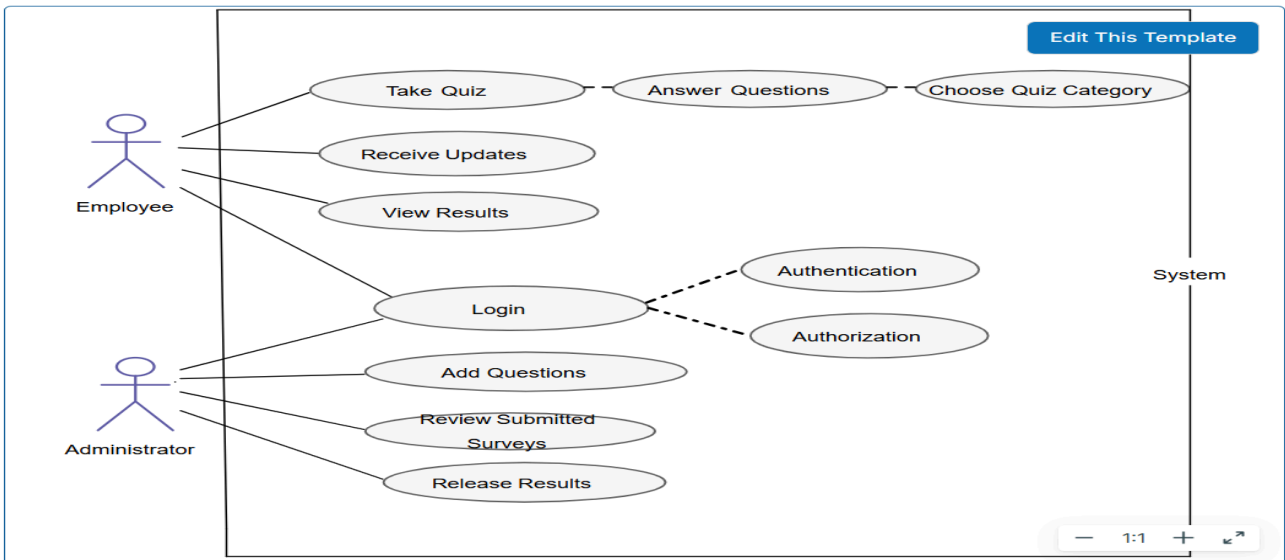
The estimation/story points were created successfully for the Online Quiz System using Poker Estimation.

EXP NO: 6	DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE
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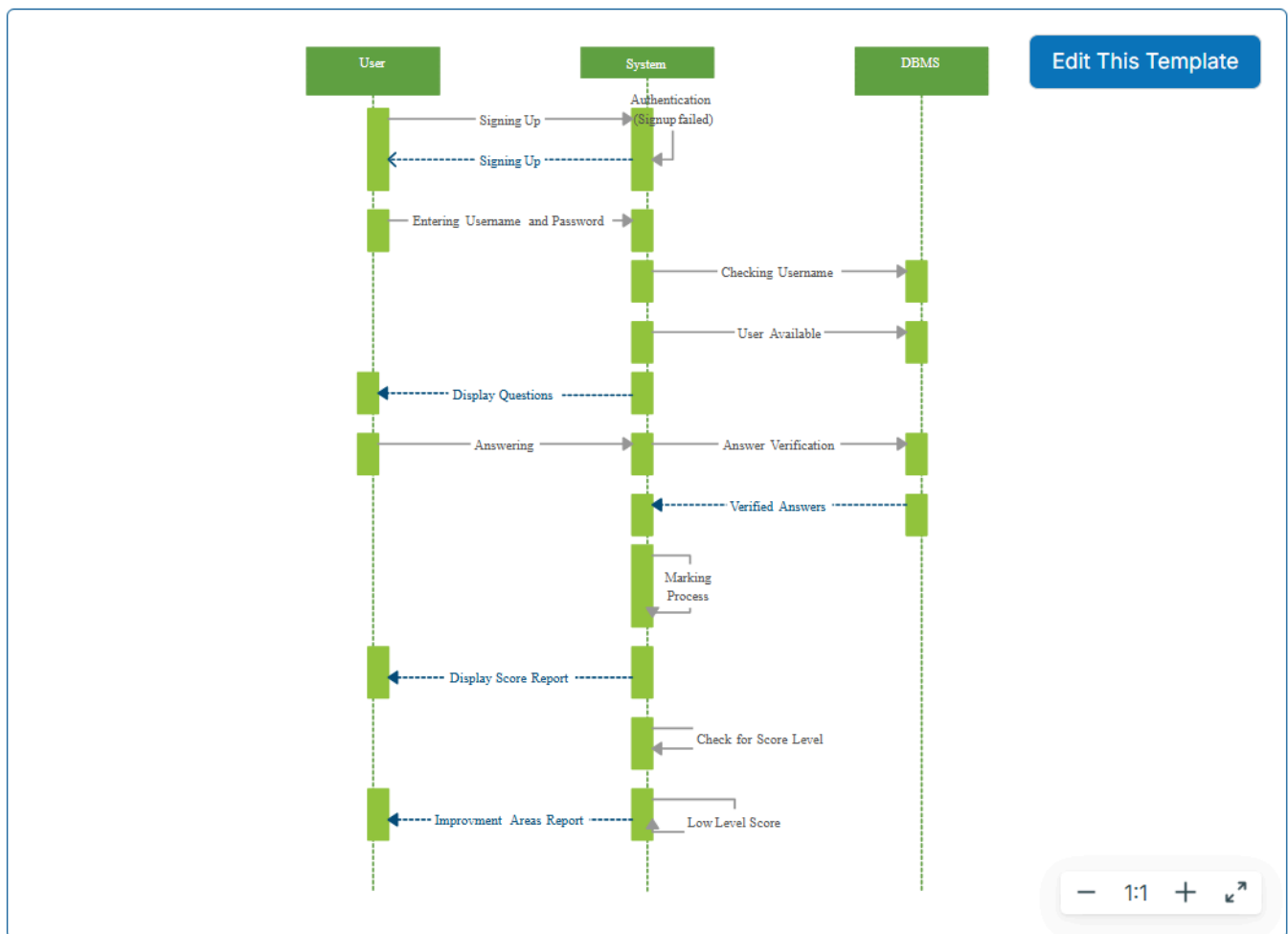
Aim:

To design a Class Diagram and Sequence Diagram for the Online Quiz System.

6A. Class Diagram



6B. Sequence Diagram



Result:

The Class Diagram and Sequence Diagram were designed successfully for the Online Quiz System.

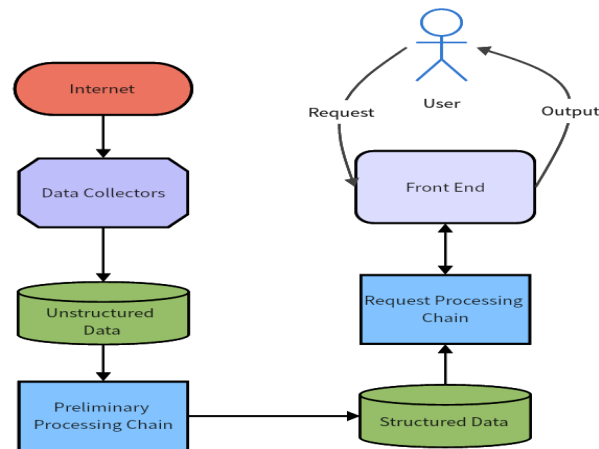
EXP NO: 7

DESIGNING ARCHITECTURAL AND ER DIAGRAMS FOR PROJECT STRUCTURE

Aim:

To design an Architectural Diagram and ER Diagram for the Online Quiz System.

7A. Architectural Diagram



7B.ER Diagram



Result:

The Architectural Diagram and ER Diagram were designed successfully for the Online Quiz System.

EXP NO: 8	TESTING – TEST PLANS AND TEST CASES
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Aim:

To create test plans and test cases for the Online Quiz System showcasing happy path and error scenarios in Azure DevOps.

Test Planning and Test Case Design Procedure

1. Understand Core Features of the Application

- User Registration and Login
- Secure Authentication System
- Online Quiz Attempt Functionality
- Quiz Timer and Auto Submission
- Score Calculation and Result Display
- Error Handling (invalid login, empty submission, server issues)

2. Define User Interactions

- Each test case represents real user behaviour such as:
Registering an account, Logging into the system, Attempting quizzes, Submitting answers, Viewing quiz results.

3. Design Happy Path Test Cases

- Focus on validating correct system behaviour under normal conditions.
- Example: User enters valid login credentials and accesses the quiz dashboard successfully.

4. Design Error Path Test Cases

- Validate system behaviour under invalid or unexpected conditions.
- Example: User enters invalid login credentials → error message displayed, Empty quiz submission → warning message shown.

5. Break Down Steps and Expected Results

- Each test case includes:
Step-by-step actions (e.g., open application → enter login credentials → click login),
Expected result (e.g., user dashboard displayed successfully)

- Ensures clarity for manual testers and automation scripts.

6. Use Clear Naming and IDs

- Test cases are uniquely identified for easy tracking:
TC01 – Successful Login Validation
TC02 – Invalid Login Error Handling
- Helps map test cases to user stories and requirements.

7. Separate Test Suites

- Test cases are grouped based on functionality:
User Authentication, Quiz Management, Result Processing, Error Handling & Edge Cases
- Improves organization and test execution flow in Azure DevOps.

8. Prioritize and Review

- Critical user actions are marked high-priority.
- Reviewed for completeness and traceability against feature requirements.

1. New test plan

The screenshot shows the 'New Test Plan' form in Azure DevOps. The form is titled 'New Test Plan' and has three main input fields: 'Name' with the value 'Online Quiz Test Plan', 'Area Path' with the value 'Online Quiz', and 'Iteration' with the value 'Online Quiz/Iteration 1'. At the bottom right, there are 'Create' and 'Cancel' buttons. The left sidebar shows the 'Test Plans' section selected, with sub-items like 'Test plans', 'Progress report', 'Parameters', 'Configurations', 'Runs', 'Exploratory sessions', and 'Artifacts'.

2. Test suite

Online Quiz

Overview

Boards

Repos

Pipelines

Test Plans

Test plans

Progress report

Parameters

Configurations

Runs

Exploratory sessions

Artifacts

Project settings

Online Quiz Test Plan

May 11 - May 18

Current

Test Suites

Filter suites by name

Online Quiz Test Plan

User Authentication Testing

New Suite

Assign configurations

Export

Open

Assign testers to run all tests

Rename

Delete

Import test suites

Static suite

Requirement based suite

Query based suite

Search

Help

User Authentication Testing (ID: 16)

Define

Execute

Chart

Static suite

Requirement based suite

Query based suite



Add a test case

Use this tab to collate, add and manage test cases

New Test Case

3. Test case

Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Weather App – Test Plans

USER STORIES

- As a user, I want to register an account so that I can attend quizzes.
- As a user, I want to log in securely to access quizzes.
- As a user, I want to attempt quizzes online.
- As a user, I want to view my score after quiz submission.
- As an admin, I want to create and manage quizzes.
- As a user, I want to see an error message for invalid login credentials.

Test Suites:

TS01 – Online Quiz Feature Testing

1. TC01 – Successful Weather Search

- **Action:**
 1. Open Online Quiz System
 2. Enter valid username and password
 3. Click Login
- **Expected Results:**
 1. Login successful
 2. Dashboard displayed
 3. User redirected to quiz page
- **Type:** Happy Path

2. TC02 – Quiz Attempt Functionality

Action:

1. Login to system
2. Select a quiz
3. Answer questions
4. Submit quiz

○ **Expected Results:**

1. Quiz submitted successfully
2. Score calculated automatically
3. Result displayed

- **Type:** Happy Path

3.TC03 – Quiz Timer Validation

- o **Action:**
 1. Start quiz
 2. Wait for timer completion
- o **Expected Results:**
 1. Quiz auto-submits when timer ends
 2. Answers saved successfully
- o **Type:** Happy Path

4.TC04 – Invalid Login Error

- o **Action:**
 1. Enter incorrect credentials
 2. Click Login
- o **Expected Results:**
 1. Error message displayed
 2. Login denied
 3. Application does not crash
- o **Type:** Error Scenario

5.TC05 – Empty Quiz Submission

- o **Action:**
 1. Start quiz
 2. Submit without answering questions
- o **Expected Results:**
 - Warning message displayed
 - Quiz handled properly
- o **Type:** Error Scenario

Test Cases

TEST CASE 18

18 Successful Login Validation

Harshith Kumar Suresh

0 Comments Add Tag

Save and Close

Follow

0 0

State Design

Area

Online Quizz

Reason New

Iteration

Online Quizz\Iteration 1

Updated by Harshith Kumar Suresh: 6m ago

Steps

Summary

Associated Automation

0

0

0

Steps

📎 📎 📎 | ↑ ↓ ✖ | @ | | B / U

Steps	Action	Expected result	Attachments
1.	Open Online Quiz System	Login successful	
2.	Enter valid username and password	Dashboard displayed	
3.	Click Login	User redirected to quiz page	

Click or type here to add a step

Recent test results

Run manual tests with web runner. [Learn how to get started](#)

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add link

[Add an existing work item as a parent](#)

TEST CASE 19

19 Invalid Login Error

Harshith Kumar Suresh

0 Comments Add Tag

Save and Close

Follow

0 0

State Design

Area

Online Quizz

Reason New

Iteration

Online Quizz\Iteration 1

Updated by Harshith Kumar Suresh: 5m ago

Steps

Summary

Associated Automation

0

0

0

Steps

📎 📎 📎 | ↑ ↓ ✖ | @ | | B / U

Steps	Action	Expected result	Attachments
1.	Enter invalid username/password		
2.	Click Login	Login denied	
3.			

Click or type here to add a step

Recent test results

Run manual tests with web runner. [Learn how to get started](#)

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

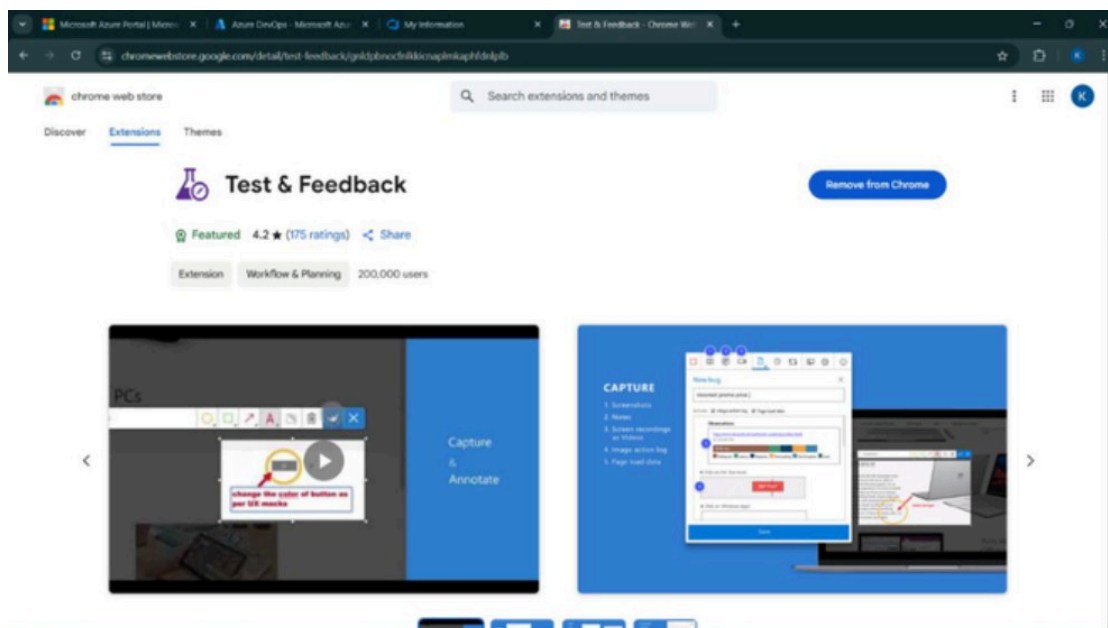
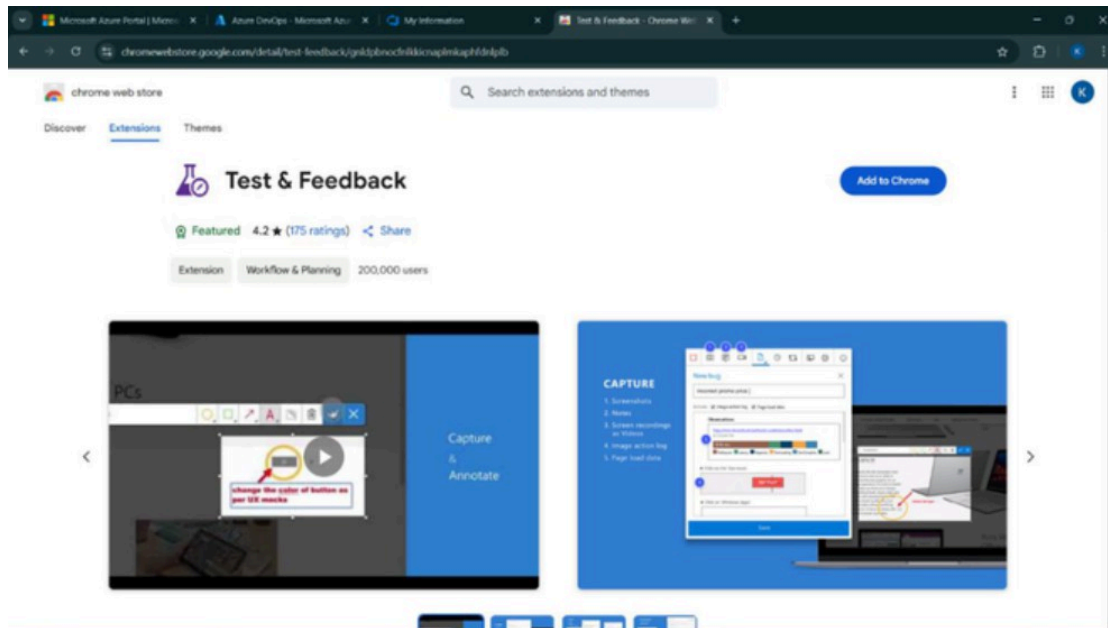
Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add link

[Add an existing work item as a parent](#)

4. Installation of test



Test and feedback
Showing it as an
extension

dev.azure.com/harshithkumars2024it/Online%20Quiz/_testPlans/execute?planId=14&suiteId=16

Azure DevOps harshithkumars2024it / Online Quiz / Test Plans / Online Quiz Test Plan

Online Quiz

- Overview
- Boards
- Repos
- Pipelines
- Test Plans**
 - Test plans**
 - Progress report
 - Parameters
 - Configurations
 - Runs
 - Exploratory sessions
 - Artifacts
- Project settings

Online Quiz Test Plan

May 11 - May 18 **Current**

0% run. [View report](#)

Test Suites

Filter suites by name

Online Quiz Test Plan

User Authentication Testing (3)

User Authentication Testing (ID: 16)

Define **Execute** Chart

Test Points (3 items)

Title	Outcome	Order	Test Case Id
☒ Successful Login Validation	Active	2	18
☐ Invalid Login Error	Active	3	19
☐ Quiz Attempt Validation	Active	4	20

Extensions

Full access
These extensions can see and change information on this site.

- NeoExamShield
- Test & Feedback
- Manage extensions

5. Running the test cases

Azure DevOps harshithkumars2024it / Online Quiz / Test Plans / Online Quiz Test Plan

Online Quiz

- Overview
- Boards
- Repos
- Pipelines
- Test Plans**
 - Test plans**
 - Progress report
 - Parameters
 - Configurations
 - Runs
 - Exploratory sessions
 - Artifacts
- Project settings

Online Quiz Test Plan

May 11 - May 18 **Current**

0% run. [View report](#)

Test Suites

Filter suites by name

Online Quiz Test Plan

User Authentication Testing (3)

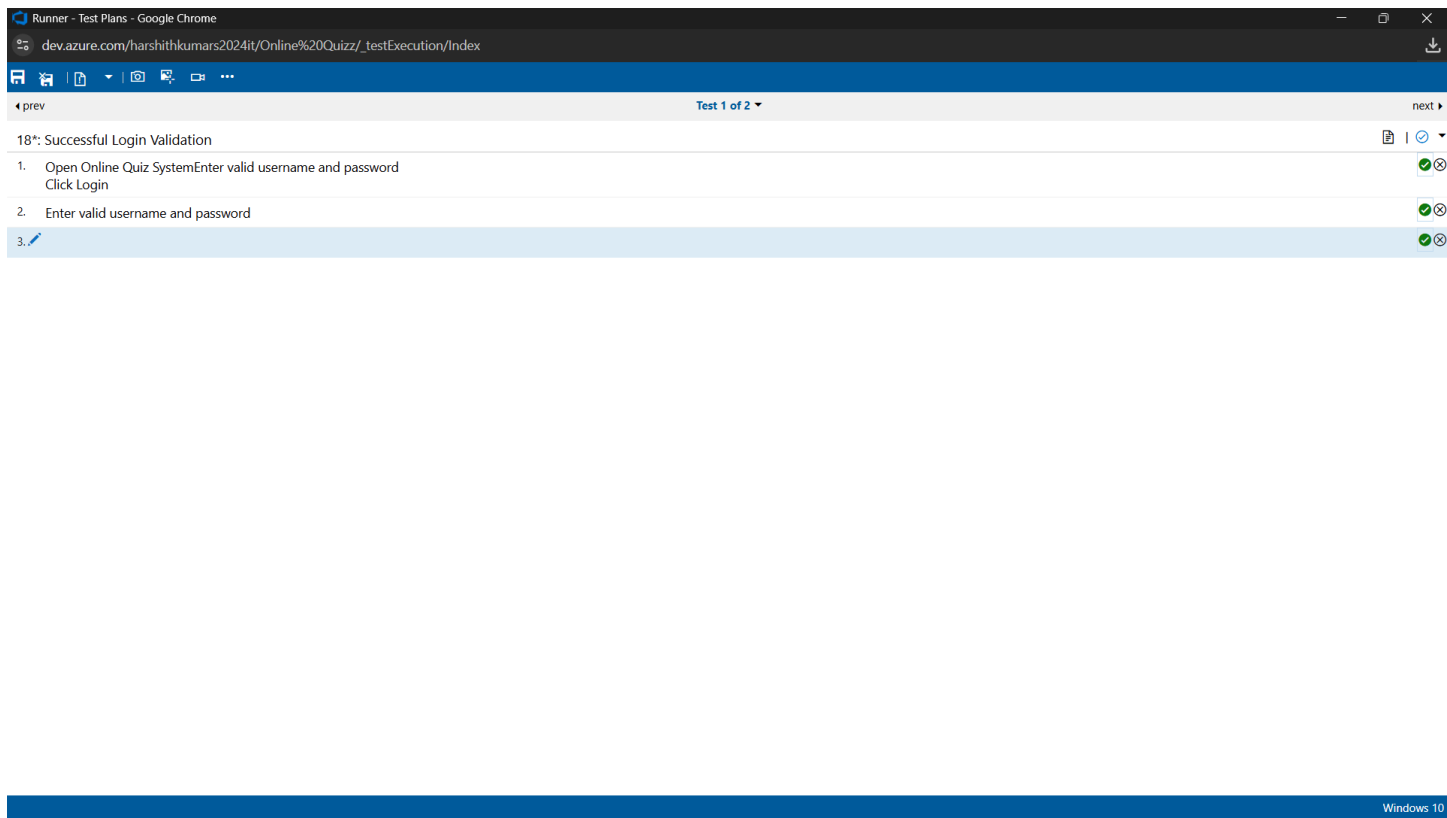
User Authentication Testing (ID: 16)

Define **Execute** Chart

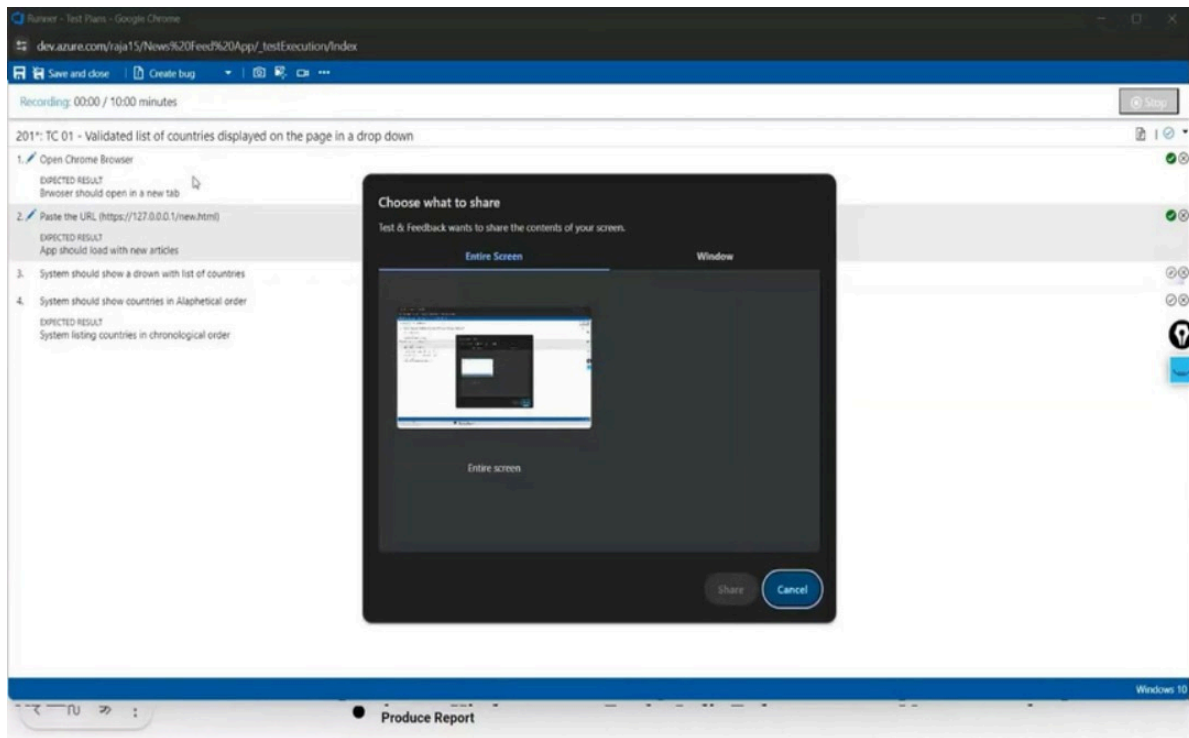
Test Points (3 items)

Title	Outcome	Order	Test Case Id
☒ Successful Login Validation	In Progress	2	18
☐ Invalid Login Error	In Progress	3	19
☐ Quiz Attempt Validation	In Progress	4	20

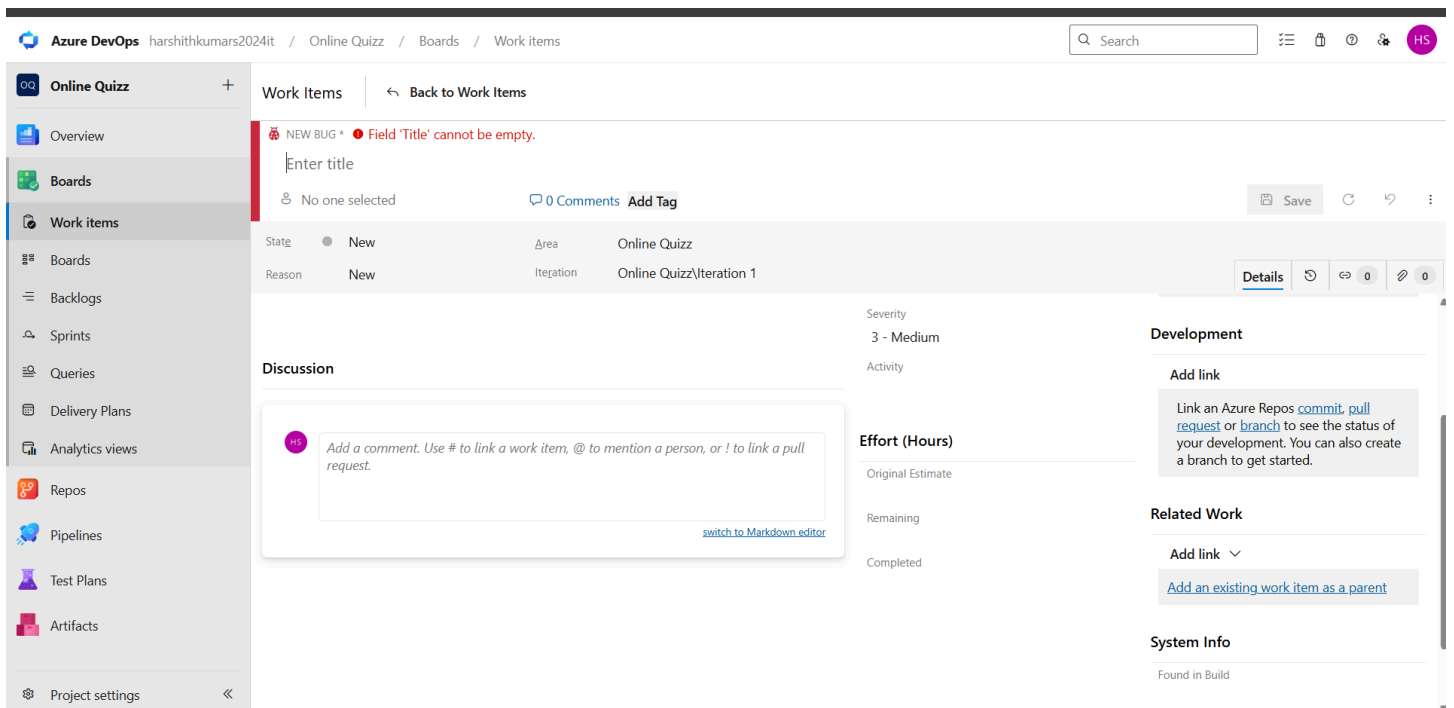
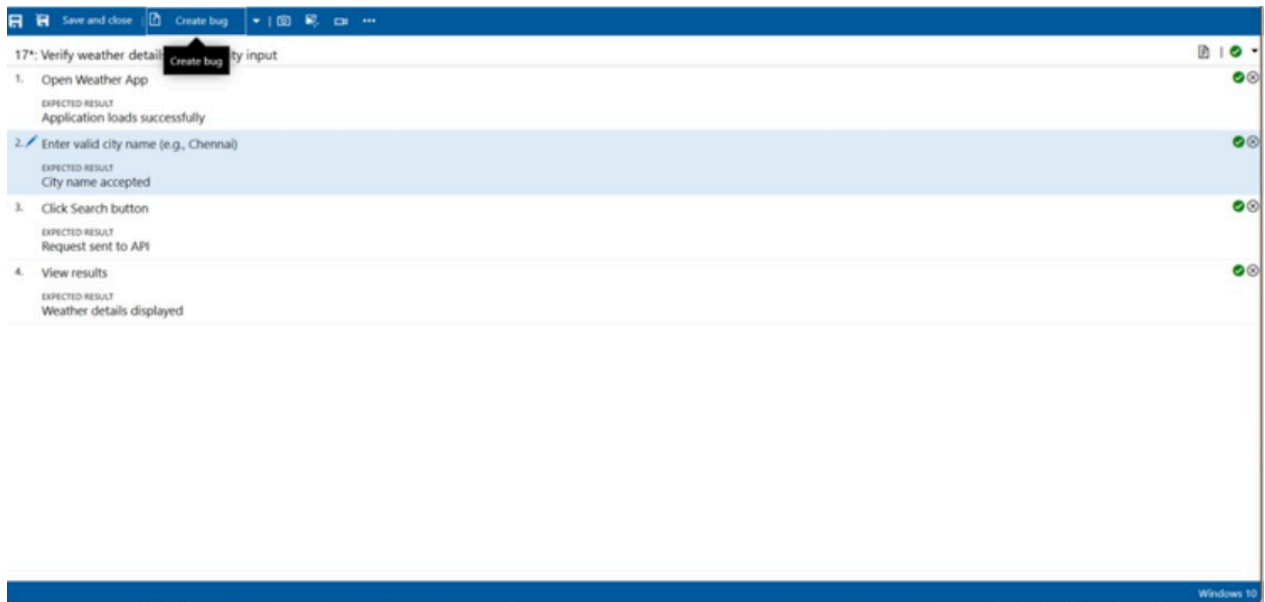
Run for web application



6. Recording the test case



7. Creating the bug



8. Test case results

The screenshot shows the Azure DevOps Test Plans interface. The left sidebar contains navigation options: Overview, Boards, Repos, Pipelines, Test Plans (selected), Test plans, Progress report, Parameters, Configurations, Runs, Exploratory sessions, and Artifacts. The main area displays the 'User Authentication Testing (ID: 16)' test plan. The 'Execute' tab is active, showing a table of test points. The test plan is for 'Online Quiz Test Plan' and is currently 'Current'. It shows a 100% run and 66% passed status. The test points table lists three items, all of which passed.

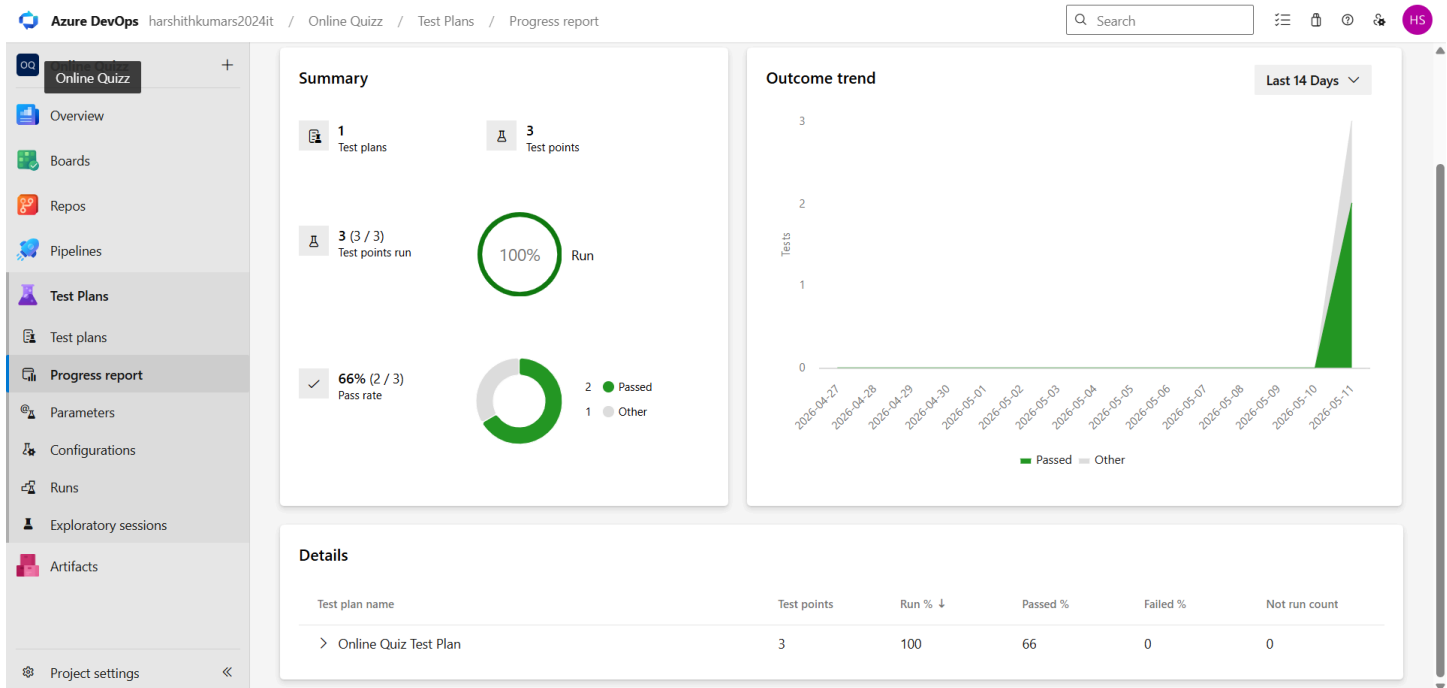
Title	Outcome	Order	Test Case Id
Successful Login Validation	Passed	2	18
Invalid Login Error	Passed	3	19
Quiz Attempt Validation	Passed	4	20

9. Test report summary

- Assigning bug to the developer and changing state

The screenshot shows the Azure DevOps Boards interface. The left sidebar contains navigation options: Overview, Boards (selected), Work items, Backlogs, Sprints, Queries, Delivery Plans, Analytics views, Repos, Pipelines, Test Plans, and Artifacts. The main area displays a bug report for 'Invalid Login Error Bug' (ID: 21). The bug is assigned to Harshith Kumar Suresh and is in the 'Active' state. The 'Repro Steps' section is empty. The 'System Info' section shows the area as 'Online Quiz' and the iteration as 'Online Quiz\Iteration 1'. The 'Discussion' section is empty. The 'Planning' section shows the resolved reason, story points, priority, severity, and activity. The 'Effort (Hours)' section shows the original estimate, remaining, and completed hours. The 'Deployment' section provides instructions on how to track releases associated with the work item. The 'Development' section provides instructions on how to link an Azure Repos commit, pull request, or branch to the work item. The 'Related Work' section is empty.

10. Progress report



11. Changing the test template

Azure DevOps harshithkumars2024it / Settings / Process

Q Search

Organization Settings

harshithkumars2024it

Search Settings

General

Overview

Projects

Users

Billing

Global notifications

Usage

Extensions

Microsoft Entra

Security

Security overview

Policies

Permissions

Boards

Process

Pipelines

All processes

Help

Filter by process name

Name	Description	Team projects
Basic (default)	This template is flexible for any process and great for teams getting started with Az...	0
Agile	This template is flexible and will work great for most teams using Agile planning me...	1
Scrum	This template is for teams who follow the Scrum framework.	0
CMMI	This template is for more formal projects requiring a framework for process improv...	0

12. View the new test case template

The screenshot displays the Azure DevOps interface for viewing a new test case template. The top navigation bar shows the user 'harshithkumars2024it' and the current path 'All processes > Agile > Test Case'. A search bar and utility icons are on the right. The left sidebar, titled 'Organization Settings', lists categories like General, Security, Boards, and Pipelines, with 'Process' selected under Boards. The main content area has a breadcrumb 'All processes > Agile > Test Case' and tabs for 'Layout' and 'States'. Below these are buttons for 'New field', 'New group', 'New page', 'Get extensions', and a refresh icon. A tabbed interface shows 'Steps', 'Summary', and 'Associated Aut...'. The 'Steps' tab contains a 'Text (multiple lines)' field. To the right, a vertical list of sections includes 'Recent test results', 'Deployment', 'Development', 'Related Work', 'Status', 'Priority', and 'Automation status'. A large dashed box on the right is labeled 'Add a field ...'.

Result:

The test plans and test cases for the user stories were created successfully in Azure DevOps with Happy Path and Error Path scenarios for the Online Quiz System.

EXP NO: 9	LOAD TESTING AND PIPELINES
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Aim:

To create an Azure Load Testing resource and run load testing for the Online Quiz System and demonstrate Azure DevOps pipelines for automated builds and deployment.

Load Testing

The Online Quiz System was tested under multiple concurrent users to evaluate:

- Server response time
- Quiz submission handling
- Database performance
- User scalability

Steps to Create an Azure Load Testing Resource:

Before you run your first test, you need to create the Azure Load Testing resource:

1. Sign in to Azure Portal
Go to <https://portal.azure.com> and log in.
2. Create the Resource
 - o Go to *Create a resource* → Search for “Azure Load Testing”.
 - o Select Azure Load Testing and click Create.
3. Fill in the Configuration Details
 - o *Subscription*: Choose your Azure subscription.
 - o *Resource Group*: Create new or select an existing one.
 - o *Name*: Provide a unique name (no special characters).
 - o *Location*: Choose the region for hosting the resource.
4. (Optional) Configure tags for categorization and billing.
5. Click Review + Create, then Create.
6. Once deployment is complete, click Go to resource.

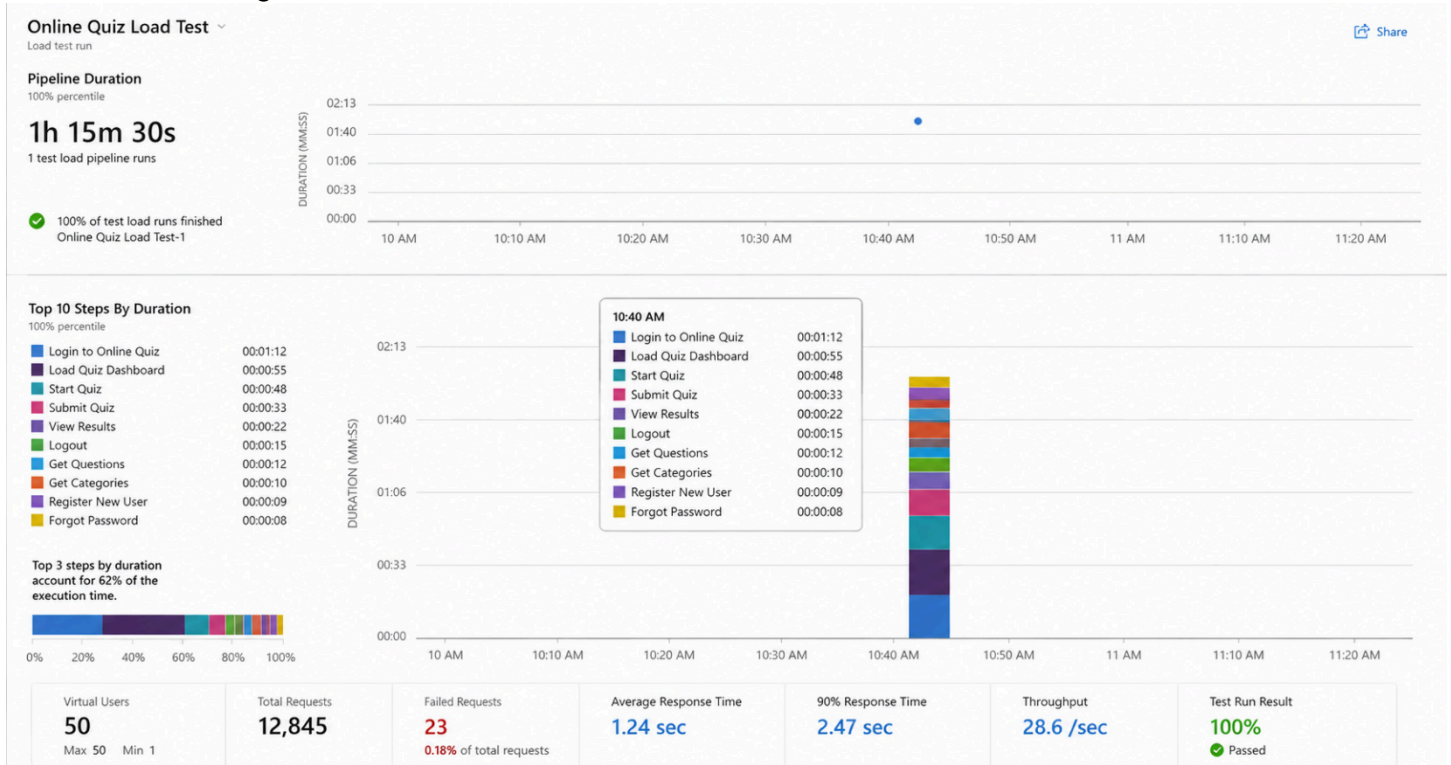
Steps to Create and Run a Load Test:

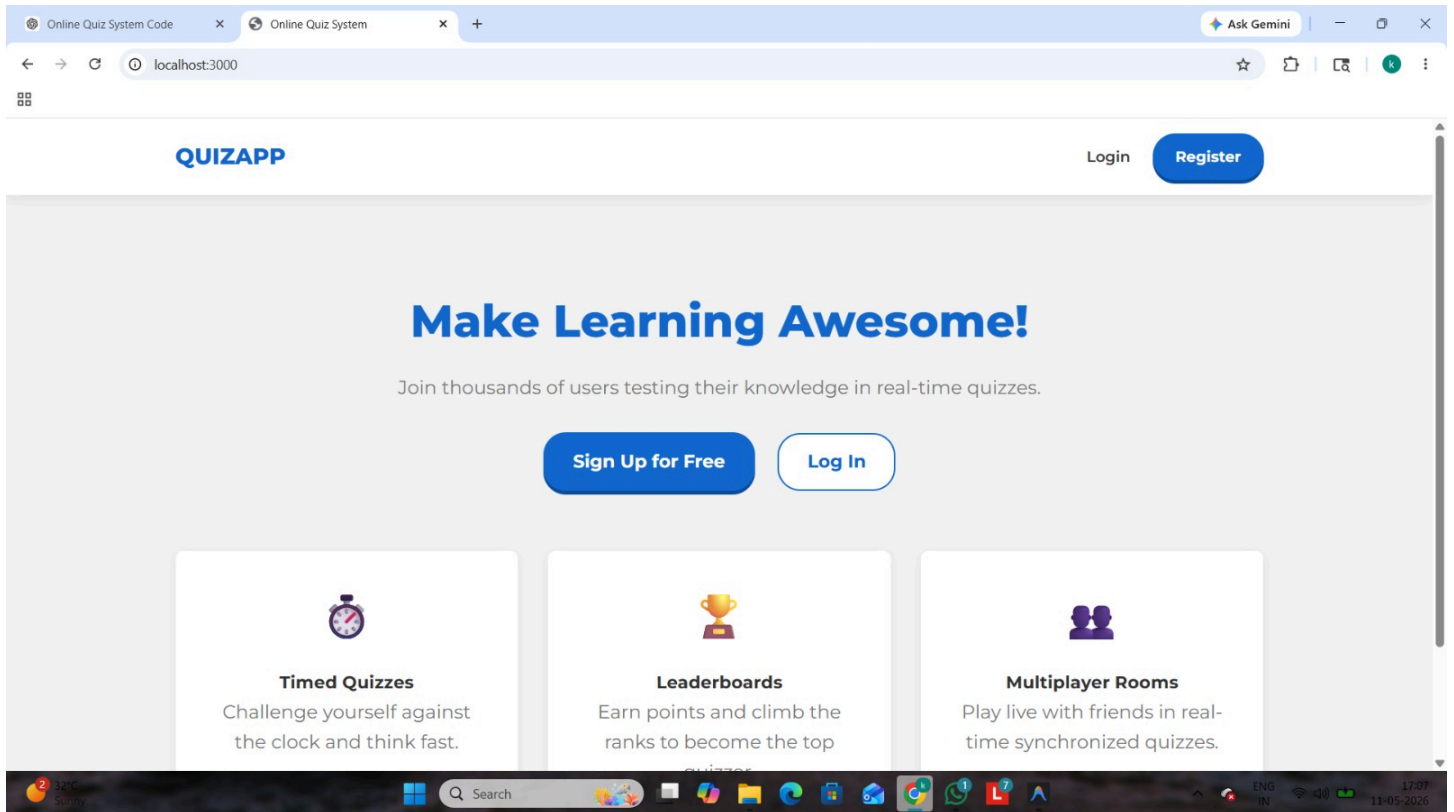
Once your resource is ready:

1. Go to your Azure Load Testing resource and click Add HTTP requests > Create.
2. Basics Tab
 - o *Test Name*: Provide a unique name.
 - o *Description*: (Optional) Add test purpose.
 - o *Run After Creation*: Keep checked.
3. Load Settings
 - o *Test URL*: Enter the target endpoint (e.g., <https://yourapi.com/products>).

4. Click Review + Create → Create to start the test.

Load Testing





Pipelines

Description:

This experiment demonstrates how to connect the GitHub-hosted Online Quiz System project with Azure DevOps. The pipeline automates project validation, build verification, and deployment tasks.

Steps:

1. Connect GitHub to Azure DevOps:
 - o In Azure DevOps, create a new project.
 - o Create a pipeline and select GitHub as the source.
 - o Authorize access to your GitHub repository, ensuring that Azure DevOps can pull the repository for your pipeline.
2. Create azure-pipelines.yml in Your Repo Root:
 - o In your GitHub repository, create a new file called azure-pipelines.yml in the root directory.
 - o Add the following basic pipeline configuration for HTML, CSS, and JavaScript-based Weather Application:

yml Code:

trigger:

- main

pool:

vmImage: 'ubuntu-latest'

steps:

- script: echo "Online Quiz System Build Successful!"
 displayName: 'Run Build Check'

- script: |
 echo "Checking project files..."
 ls
 displayName: 'List Project Files'

3. Pipeline Tasks Include:
 - o Setting up the pipeline environment using the default Azure-hosted agent (ubuntu-latest).
 - o Validating the Online Quiz project files stored in the repository.
 - o Running a simple build verification script to ensure the pipeline is configured correctly.
 - o Listing project files to confirm successful repository integration and pipeline execution.
4. Run and Monitor Pipeline:
 - o Commit changes to the main branch of your repository to trigger the pipeline in Azure DevOps.
 - o Monitor the pipeline execution logs in the Azure DevOps portal to verify build steps, detect errors, and confirm successful execution of the configured tasks.

Pipeline

The screenshot displays the Azure DevOps web interface for a pipeline named '#20260511.2 • Update azure-pipelines.yml for Azure Pipelines'. The interface includes a left-hand navigation pane with options like Overview, Boards, Repos, Pipelines, Environments, Library, Test Plans, and Artifacts. The main content area shows the pipeline's summary, including the user who manually ran it (Harshith Kumar Suresh), the repository (HarshithKumarS/Software-construction), and the commit hash (c7bd03c9). Below the summary, a 'Stages' section indicates that 1 job was completed in 18 seconds. The top right of the interface features a search bar and user profile information.

Azure DevOps harshithkumars2024it / Online Quiz / Pipelines / HarshithKumarS.Software-c... / 20260511.2

Search

#20260511.2 • Update azure-pipelines.yml for Azure Pipelines Cancel

HarshithKumarS.Software-construction

Summary Code Coverage

Manually run by HS Harshith Kumar Suresh View change Parameters

Today at 22:08

Repository and version

HarshithKumarS/Software-construction

main c7bd03c9

Related

0 work items

0 published

Tests and coverage

Get started

Stages Jobs

Stage

1 job completed 18s

Job 18s

Project settings

Azure DevOps harshithkumars2024it / Online Quiz / Pipelines / HarshithKumarS.Software-c... / 20260511.2

Search

Online Quiz +

Overview

Boards

Repos

Pipelines

Pipelines

Environments

Library

Test Plans

Artifacts

Project settings <<

← ✓ Jobs in run #20260511.2
HarshithKumarS.Software-construction

Jobs

✓ Job

✓ Job

Pool: [Azure Pipelines](#)
Image: ubuntu-latest
Started: Today at 22:08:15
Duration: 18s

View raw log

- ✓ Initialize job 2s
- ✓ Checkout HarshithKumarS.Software-construction@main 2s
- ✓ Set up job 1s
- ✓ Run echo "OnLine Quiz System Build Successful!" 2s
- ✓ Run echo "Checking project files..." 1s
- ✓ Run ls 2s
- ✓ Post-job: Checkout HarshithKumarS.Software-construction@main 1s
- ✓ Finalize Job 1s
- ✓ Report build status 1s

Job completed successfully

Result:

Successfully created the Azure Load Testing resource and executed load testing for the Online Quiz System and demonstrated Azure DevOps pipelines successfully.

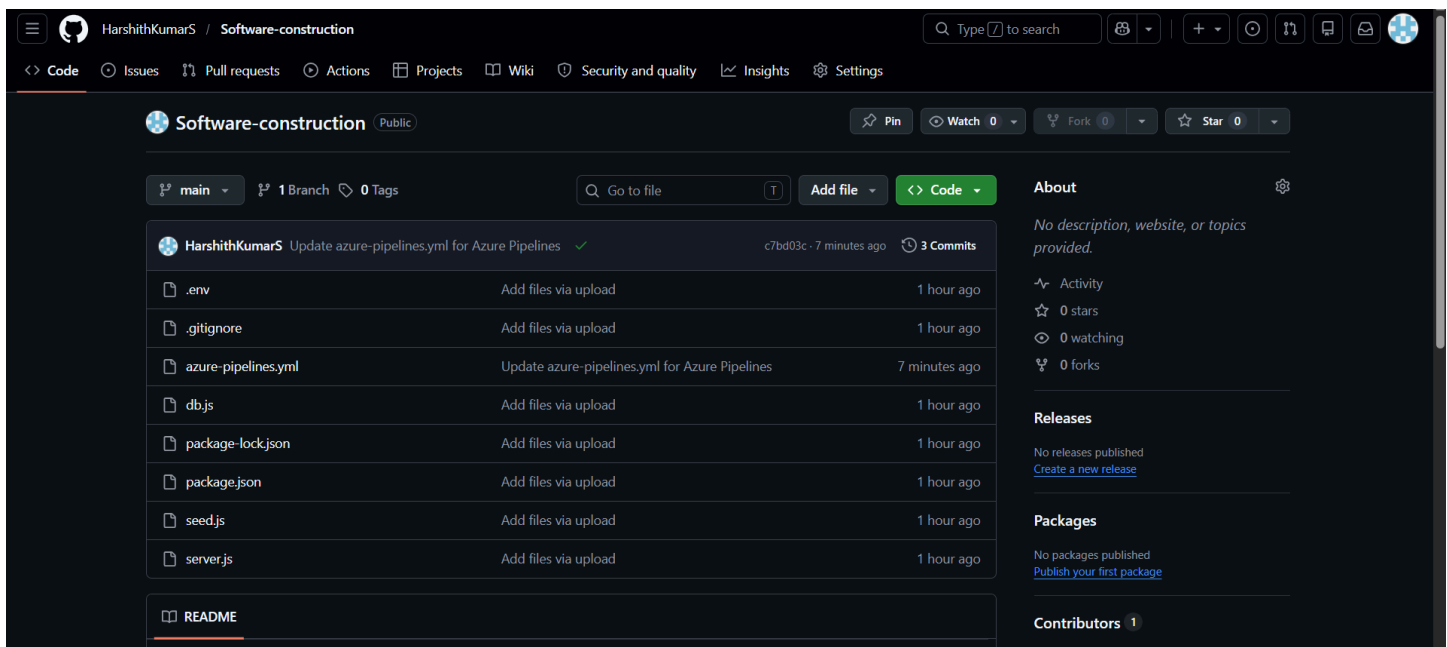
EXP NO: 10

GITHUB: PROJECT STRUCTURE & NAMING CONVENTIONS

Aim:

To provide a clear and organized view of the Online Quiz System project structure and file naming conventions for easy navigation and maintenance.

GitHub Project Structure



Result:

The GitHub repository clearly displays the organized project structure and consistent naming conventions for the Online Quiz System project.